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Task Description:	This NASA Specialized Center of Research (NSCOR) aims to model the relative risk of colonic and stomach tumorigenesis for high priority space radiation beams and compare to gamma radiation where human epidemiologic data are available. It will rely primarily on mouse models of gastrointestinal (GI) cancer and human colonic epithelial cell (HCEC) approaches that have been optimized for low dose exposures. This will involve quantitative and qualitative analyses of tumors as well as normal GI epithelium after radiation. Results will be extrapolated to radiation-induced molecular changes in state-of-the-art 2D and 3D cultured human colonocytes. While limited studies at select loci, such as HRPT, have shown high energy (HZE)-induced mutations including splicing errors, frameshifts, and complex changes, we will carry out genome-wide analysis using the power of modern nextgen sequencing (NGS) to assess the spectrum of mutations in radiation-induced tumors, and also persistent genome-wide epigenomic changes in normal cells after radiation. These NGS omics findings will be integrated with transcriptomics and metabolomics results as well as events at the protein and cellular levels to develop a systems biology model for GI tumorigenesis. By assessing both quantitative in vivo tumorigenesis data and molecular events in mouse tissues and human colonocytes, space radiation-induced tumorigenesis will be compared to human epidemiologic data for low LET (linear energy transfer) radiation exposures, and approaches will be developed to model human risks for GI cancer by space radiation. We continue to employ two models of GI cancers: the APC1638N/+ model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa1638. These mice develop very few adenomas in the small as well as in the large intestine but post-radiation tumor numbers show marked increases. Radiation also induced substantial gastric tumorigenesis as well in this model. The second model is the CDX2P Apc flox/+ that has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequently a low rate of adenocarcinomas in the colon that is increased post-irradiation. Data being collected from these two relevant mouse models will be used to calculate risk estimates for NASA. Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation compared to the standard risks based on atomic bomb survivors and other exposure populations. In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, in vitro human cell culture and animal models for colorectal cancer (CRC) initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer-related endpoints and systems biology (genomics, proteomics, and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The studies of low- and high-LET radiation on colonocytes gain more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation. Plans for this proposal build on our past accomplishments with GI tumorigenesis models and are focused on studies at low doses with an expanded range of NASA's high priority beams. This proposal is also focused on assessing the effects of gender, dose-rate, age, and genetic heterogeneity in mouse models. Molecular events triggered by space radiation will be investigated in depth and the mechanistic basis for persistent aberrant cell-signaling after HZE radiation will be dissected. Genome-wide NGS combined with other omics approaches will contribute to a systems biology understanding of the effects of space radiation in the intestines and stomach. Both quantitative murine tumor data and results for underlying molecular events in mouse and human GI cells will contribute to modeling of human risks for GI cancer by space travel. The overall aims of this NSCOR are summarized below: Aim 1 (Project 1). Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis. Aim 2 (Project 2). Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation Aim 3 (Project 3): Characterization of IR-induced neoplastic events in normal diploid human colonocytes Aim 4 (Project 4). Development of systems biology (Project 4A) and mathematical modeling (Project 4B) approaches for GI cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S, and American Cancer Society estimates that there will be 140,250 new cases of colorectal cancer (CRC) with 50,630 persons predicted to die of the disease in 2018. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and colorectal carcinogenesis.
	The overall plan is to apply the well-characterized APC mouse models (APC1638N/+ and CDX2P Apcflox/+) of colorectal cancer (CRC) for development of risks estimates for various radiation exposures during manned space flight. We already know from studies in atomic bomb survivors that radiation is risk factor for CRC. Space radiation with its novel characteristics is expected to have greater risk relative to radiation on Earth. A standard approach in radiation biology towards risk estimation is to determine relative biologic effectiveness (RBE) of various parameters for space radiation compared to terrestrial radiation exposures. However, currently, there is much uncertainty in predicting risk of CRC in astronauts undertaking long duration space missions due to lack of sufficient in vivo mechanistic data in human or in animal models representing the human disease. Considering the limitation of acquiring human data, our study is focused on acquiring sufficient data in mouse models of human CRC with an aim to aid in risk model development for safe human space exploration. Therefore, qualitative and quantitative data on intestinal tumorigenesis in mice generated in this project will be used for quantitative dose response analyses. While such analyses will provide an estimate of risk in mouse models, our ultimate aim is to combine mouse model data from space radiation exposure and human epidemiological data from atomic bomb survivors to develop mathematical risk prediction models of CRC for astronauts.

Task Progress:

We have continued irradiation of the APC1638N/+ mouse model to obtain quantitative and qualitative tumorigenesis data at lower doses of space radiation beams spanning a range of LET and energies. We participated in two beam runs in the spring and fall of 2017 and we are breeding and genotyping mice for the spring 2018 run at NASA Space Radiation Laboratory (NSRL). These beam runs have allowed us to irradiate mice for fulfilling statistically required numbers decided in consultation with our Project 4B lead Dr. Brenner at Columbia University. We are also assessing the effect of chronic aspirin administration on tumor frequency; there are human data indicating that low-dose aspirin can have a protective effect against CRC. Our previous data in APC mutant mouse model demonstrated higher intestinal tumorigenesis after exposure to 12C, 28Si, and 56Fe radiation relative to gamma rays. We also observed important qualitative differences between heavy ions and gamma rays in both normal GI epithelial cells and tumors. However, our lowest dose for these experiments was 10 cGy. This year we exposed APC1638N/+ male and female mice to 5 cGy of gamma-rays, 16O (325 MeV/n), 4He (250 MeV/n), 28Si (300 MeV/n), and 56Fe (1000 MeV/n) with an aim to assess tumorigenesis at doses lower than the previous ones with a broader range of ions. We also exposed mice to 10 cGy of 16O (325 MeV/n) and 4He (250 MeV/n), which are new beams proposed in the renewed NSCOR, to get statistically required numbers. In an effort to assess effects of genetic heterogeneity on space radiation-induced intestinal tumorigenesis, we performed backcrossing of APC1638N/+ in C57BL/6J background into C3H/HeN and BALB/c background. The backcrossing was done for 10 generations and Sequencing of the region spanning the APC gene locus after backcrossing for 10 generations indicate that much of the chromosomal DNA flanking this locus is now either from C3H/HeN or BALB/c and not from C57BL/6J. We have scheduled beam time in 2018 fall to irradiate APC1638N/+ mice in C3H/HeN and BALB/c background to compare tumorigenesis in these vs. C57BL/6J background.

We expanded our GCR simulations (GCRsim) using both colon and lung cancer susceptible mouse model in comparison to wild type mice. In our initial experiments we used the following beams: protons (1H), helium (4He) and silicon (28Si). We used fast switching of these beams at the NSRL using a dose rate of 0.5 cGy/min and the large 60 x 60 beam. The total dose mice were exposed to was either 30 or 27 cGy. We used 20 cGy of proton (120Mev/n), 5 cGy of helium (250 MeV/n) and either 2 or 5 cGy of silicon (300 MeV/n). We previously showed that 30 cGy of proton and 5 cGy of silicon as single beam exposures did not increase polyp size or number of polyps in the CPC;APC mouse model. However, we observed 100 days post-IR that the number and polyp sizes were increased in mice exposed to the GCRsim in comparison to protons alone or unirradiated mice. Initial results indicate differential expression in normal colonic tissue in the CPC;APC mice 100 days after GCRsim. These included increases in senescence-associated inflammatory response genes and an increase in senescence markers (such as p16INK4a). We also varied the order of the different ions. While the trend was similar irrespective of beam order, GCRsim that begin with protons and end with silicon showed higher expression of specific inflammatory genes and larger and more polyps compared to unirradiated or proton only irradiated mice. These inflammatory genes have been previously shown to correlate with poorer overall survival in humans with colon cancer. We are currently testing a model of why the order of irradiation matters. In addition, we have completed a small pilot series of experiments to test aspirin or CDDO (an anti-oxidant/anti-inflammatory agent that is orally available and currently in clinical development) as effective countermeasures of IR-induced gene expression changes and tumor progression. We are also comparing the results in the colon susceptible mice to a mouse model of lung cancer. Initial results indicate that mice irradiated with GCRsim at 100 days post-IR have higher levels of lipid peroxidation in the serum as well as an increased number of premalignant lung lesions.

Understanding molecular underpinning of space radiation-induced adverse effects in normal gastrointestinal (GI) tissue is a key factor for pathophysiologic risk assessment. This year we focused on assessing long-term effects of heavy ion radiation on autophagy in intestinal epithelial cells two months after exposure. Autophagy is a complex catabolic process involved in removing and recycling damaged or unwanted cellular constituents in autophagolysosomal vesicles. Reduced autophagy is known to promote cellular stress not only due to increased accumulation of damaged proteins but also due to decreased removal of damaged mitochondria (mitophagy). Our data in wild type C57BL/6J mice demonstrate that exposure to 1.6 Gy of 56Fe radiation led to decreased autophagy and increased oxidative stress suggesting that heavy ion radiation is propelling cells into persistent dysregulated homeostasis. Deregulation of intestinal homeostasis is further strengthened by the fact that there is increased PI3K/Akt signaling invariably linked to increased reactive oxygen species (ROS) production as well as Akt-mediated activation of mTOR, which is known to downregulate autophagy. The generated data is expected to contribute towards developing GI risk estimates of heavy ion radiation, which is qualitatively high linear energy transfer (high-LET) in outer space relative to gamma-rays, which are low-LET. Currently, the persistent effects studies have been expanded to additional priority beams at lower radiation doses and dose rates more relevant to the space environment. Additionally, we have focused on intestinal stem cell (ISC) studies in male Lgr5 (Lgr5-EGFP-IRES-creERT2) mice in C57BL/6J background using low doses (<1 Gy) 28Si (300 MeV/n) or 56Fe (1000 MeV/n) ions. Since we observed persistent stress in intestinal epithelial cells, we are investigating to determine if persistent stress is originating from stem cell and propagating to differentiated epithelial cells. Lgr5 mice were exposed to 0.5 Gy of 56Fe and GFP+ Lgr5 cells were sorted from isolated intestinal cell using flow cytometry. Sorted cells were pelleted by cytospin and the pellet was paraffin embedded for sectioning and immunostaining. Our stem cell data show that 56Fe radiation exposure was associated with increased oxidative stress and DNA damage in intestinal stem cells. We also show that at least some of the ISC are senescent (senescent marker Glb1 positive) suggesting the possibility of senescence associated secretory phenotype associated inflammatory stress. Future plan involves understanding whether stem cell stress is propagated to progeny and developing countermeasure strategy to block space radiation-induced persistent stem cell stress.

We have also isolated colonocytes from normal and colon cancer susceptible mice. In addition to determining the best methods to culture these cells in 2D monolayers, we have establish 3D organotypic cultures from normal colonocytes and those from colonic polyps. The cells expand in size and number in organotypic culture (Matrigel) over a week period and can be trypsinized and re-cultured in 2D or 3D again. In initial experiments we established 3D cultures from WT and from the CPC;APC mouse model and irradiated them with either low or high LET IR and then placed them back in 2D culture. In comparison we irradiated intact mice (WT and CPC;APC mice) with the same doses and then removed colonocytes and established them in 2D and 3D cultures. These cells are being passaged and tested for cancer associated changes. In other experiments we initiated experiments to compare the CPC;APC mouse model to other cancer susceptible models of cancer such as a lung cancer mouse model. Initial results indicate that human bronchial epithelial cells irradiated in 2D culture form more colonies in soft agar compared to 3D-irradiated cells. These results indicate that estimates of cancer progression based on 2D culture may overestimate the cancer risks. Going forward, we plan to test a biological countermeasure, CDDO, to determine if the changes observed can be reduced or prevented by either providing CDDO prior to or after a GCR simulation.

	The goal of the genomics project is to define the features of the tumor genomes arising in our murine models after gamma, proton, and HZE exposure. Using our well established next generation sequencing pipeline, we used a targeted approach to sequence the Apc and Trp53 regions to establish tumor purity and to identify the best samples for larger scale analysis. At the same time, we can identify any mutations occurring in these regions. Single nucleotide polymorphism analysis of the Apc region from a representative group of 12 tumors and 7 normal adjacent tissues was performed. Allelic imbalance is observed in all tumors, but there is considerable variation in tumor purity. A total of 34 tumors from the Fornace and Shay labs were sequenced. In this fashion, we have identified tumors with the least dilution by normal DNA that are most suited to large scale sequencing. We are currently accessioning new samples in order to have sufficient material to test the hypothesis that HZE radiation has a recognizable mutation signature. From targeted sequencing, it appears that loss of the entire region is the primary mechanism of tumor initiation under all conditions. No single nucleotide mutations, small INDELS or structural rearrangements in or flanking Apc have been identified. Excellent, whole genome copy number profiles can also be derived from this data. Astronauts performing space exploration missions will be exposed to heavy ion radiation, which can be considerably more carcinogenic per unit dose than terrestrial radiations such as gamma-rays or x-rays. Our project within the space radiation and gastrointestinal cancer NSCOR focuses on developing biologically based mathematical models to predict heavy ion induced cancer risks at doses and exposure durations relevant for space missions. Specifically, our modeling efforts are aimed at addressing the following important question: do the risks from protracted heavy ion exposures in space differ from the risks of such irradiation delivered over a short time in terrestrial experiments? We proposed a model of heavy ion carcinogenesis and fitted it to radiation lung carcinogenesis data in humans and rats. The model predicts that, at doses expected during astronaut missions, protraction of heavy ion exposure does not have a major effect on lung cancer risks. In other words, the risks from small doses of heavy ion irradiation delivered over a short time or a long time are likely to be similar. Overall, we plan to generate in vivo data on tumor frequency and mechanisms of tumor development using NASA mandated additional priority space radiation beams at lower doses relevant to space environment. These data will be used to reduce uncertainty in CRC risk prediction and develop reliable risk prediction model for humans.

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